Status	Finished
Started	Wednesday, 21 October 2026, 4:27 PM
Completed	Wednesday, 21 October 2026, 4:31 PM
Duration	3 mins 31 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is based on the divide-and-conquer approach
c. It is less efficient than bubble sort
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It avoids the need for recursion
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Dividing the list into halves
- c. Sorting the entire list in one pass
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Quick sort
- b. Selection sort
- c. Merge sort
- d. Heap sort

The correct answer is:
Merge sort

Question **5**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Complexity
- c. Sorting
- d. Merging

The correct answer is:
Complexity

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting decreases the efficiency of selection operations
- c. Sorting is rarely used in programming
- d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **7**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **8**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Quick sort
- b. Insertion sort
- c. Heap sort
- d. Bubble sort

The correct answer is:
Insertion sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Bubble Sort
c. Quick Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Merge Sort
b. Bubble Sort
c. Quick Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 200,470,80,150,40,90,300,400,70,120
c. 80,150,200,470,40,90,300,400,70,120
d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **12**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Selection
- c. Distribution
- d. Extraction

The correct answer is:
Insertion

Question **13**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It has no effect on the search operation
- b. It increases the number of comparisons needed
- c. It allows for faster searching
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are compared element by element
- b. They are split again into smaller sublists
- c. They are discarded
- d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. order()
b. sorted()
c. sort()
d. arrange()

The correct answer is: sorted()